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LINEAR FASTENER

Abstract

The present invention relates to improvements for tactile guidance for closing a linear fastener. Such fasteners are known for use on reclosable storage bags. Disclosed is a fastener assembly comprising first and second opposing flanges. Each flange can have linear male and female closure elements that interlock with matching male and female elements of the second flange. The fastener assembly can further comprise a linear arrangement of perforations between the closure elements to provide enhanced tactile feedback while the fastener is closed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] Not applicable.
BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to improved fastener closures for use with reclosable bags. Particularly, the present invention relates to fasteners with linear male and female closure elements.

2. Description of the Related Art

[0003] Most commonly, reclosable plastic bags have one or more pairs of opposing, interlocking fasteners or closures near the top opening of the reclosable bag. The closure may generally be opened and closed many times and are typically designed to ensure that the contents of the reclosable plastic bag are securely contained within the bag when the opposing fasteners are mutually engaged.

[0004] The fasteners of reclosable bags can be opened and closed in a number of different ways. For example, a slider or zipper device can be incorporated into the bag design to facilitate the engagement and disengagement of the opposing closures. However, many reclosable bags have closures that are designed to be opened by physically pulling the closures apart and closed by pressing the interlocking closure elements together with one's finger. This type of reclosable bag closure is often referred to as a press-to-close reclosable bag.

[0005] Often, press-to-close bags utilize a pair or more of cooperating interlocking closures or fasteners with each fastener comprising cooperating male and female closure elements. These multiple closures are placed relatively close to each other so that pressure from a single finger can simultaneously close both closures. U.S. Pat. Publ. No. 2011/0311167 to George M. Hall, with a filing date of Jun. 21, 2020, which is hereby incorporated by reference into this disclosure, discloses one such arrangement of multiple closures for a reclosable bag.

[0006] These multiple closures provide certain advantages, such as improved burst resistance for preventing the bag from bursting open under pressure. Such a design also permits the outer or upper closure to have indentations in one of its closure elements to provide a clicking sensation when the cooperating elements engage and thus provide feedback to the user that the closure is aligned and being properly closed.

[0007] For a bag with multiple interlocking closures, such as the closure system disclosed by Hall, it would be desirable to provide a tactile guide path to indicate the appropriate path to follow when closing the bag. Such closure assemblies are known in the art. For instance, U.S. Pat. Publ. No. 2010/0303390 to Ackerman describes various forms of linearly arranged partial indentations placed in between upper and lower fastener assemblies for providing a tactile guide path. The publication discloses that these partial indentations provide a tactile guide path for a user to follow when closing the bag. However, the partial indentations may be too subtle for providing an adequate tactile guide path.

[0008] In consideration of the shortcomings of the above discussed prior art, it would be desirable to provide a fastener assembly for reclosable bag that provides increased tactile guidance to a user when closing the bag. It would further be desirable to provide a fastener system that provides additional guidance on the direction to follow when a user is closing the bag. The present invention represents a novel solution to address these needs.

SUMMARY OF THE INVENTION

[0009] According to at least one embodiment of the present invention, a fastener can comprise a first flange and an opposing second flange. First male and female closure elements can extend distally from a first surface of the first flange and a first closure gap can be between the first male and female closure elements. Additionally, second male and female closure elements can extend distally from a first surface of the second flange. The first male closure element can interlock with the second female closure element and the first female closure element can interlock with the second male closure element. Further, a plurality of perforations can be arranged within the first closure gap to provide tactile guidance for closing the fastener. The perforations can extend through the first flange from the flange's first surface to its second surface.

[0010] In certain embodiments of the present invention, the first flange can extend from a first end to an opposing second end and the first male and female closure elements can extend between the first and second ends of the first flange. The plurality of perforations can extend between the first and second ends of the first flange and the plurality of perforations can be arranged along a single line. In addition, a first perforation of the plurality of perforations can be proximate to the first end of the first flange and a last perforation of the plurality of perforations can be proximate to the second end of the first flange. Further, one or more central perforations of the plurality of perforations can be located between the first and last perforations. The one or more central perforations can be larger than the first and last perforations or the first and last perforations can be larger than the one or more central perforations. Also, a plurality of first side perforations of the plurality of perforations can be distributed between the first perforation and the one or more central perforations and a plurality of second side perforations of the plurality of perforations can be distributed between the last perforation and the one or more central perforations. A plurality of first side perforation gaps can comprise gaps between adjacent perforations of the plurality of first side perforations and a plurality of second side perforation gaps can comprise gaps between adjacent perforations of the plurality of second side perforations.

[0011] In further embodiments of the present invention, each of the plurality of first side perforation gaps can increase as the gaps move further away from the first side perforation and towards the one or more central perforations. Each of the plurality of second side perforation gaps can increase as the gaps move further away from the last side perforation and towards the one or more central perforations. In the alternative, each of the plurality of first side perforation gaps can decrease as the gaps move further away from the first side perforation and towards the one or more central perforations and each of the plurality of second side perforation gaps can decrease as the gaps move further away from the last side perforation and towards the one or more central perforations. Further, each of the plurality of first side perforations can increase in size as the perforations move away from the first side perforation and towards the one or more central perforations and each of the plurality of second side perforations can increase in size as the perforations move away from the last side perforation and towards the one or more central perforations. In the alternative, each of the plurality of first side perforations can decrease in size as the perforations move away from the first side perforation and towards the one or more central perforations and each of the plurality of second side perforations can decrease in size as the perforations move away from the last side perforation and towards the one or more central perforations.

[0012] Additionally, in certain embodiments of the present invention, a second surface of the first flange can curve inwards for one or more of the plurality of perforations and the curved surface can be centered about a center of the one or more perforations. Furthermore, a first surface of the first flange can have an annular shaped wall surrounding one or more perforations of the plurality of perforations. And, the annular shaped wall can be jagged and irregular at its upper distal end.

[0013] In another embodiment of the present invention, a fastener can comprise a first flange and an opposing second flange. First male and female closure elements can extend distally from a first surface of the first flange and a first closure gap can be between the first male and female closure elements. Second male and female closure elements can extend distally from a first surface of the second flange. The first male closure element can interlock with the second female closure element and the first female closure element can interlock with the second male closure element. A plurality of perforations can be arranged within the first closure gap and the plurality of perforations can vary in size.

[0014] For certain embodiments of the present invention, a plurality of perforation gaps can comprise gaps between adjacent perforations of the plurality of perforations and the plurality of perforation gaps can vary in size. Further, one or more central perforations of the plurality of perforations can be located between opposing first and second ends of the first flange. A plurality of first side perforations of the plurality of perforations can be distributed between the first end of the first flange and the one or more central perforations. The plurality of first side perforations can vary in size from the one or more central perforations. Additionally, a first perforation of the plurality of perforations can be proximate to a first end of the first flange and a last perforation of the plurality of perforations can be proximate to a second end of the first flange. Lastly, the plurality of first side perforations can increase in size as the perforations move away from the first side perforation and towards the one or more central perforations.

[0015] In an additional embodiment of the present invention, a fastener can comprise a first flange and an opposing second flange. First male and female closure elements can extend distally from a first surface of the first flange and a first closure gap can be between the first male and female closure elements. Second male and female closure elements can extend distally from a first surface of the second flange. The first male closure element can interlock with the second female closure element and the first female closure element can interlock with the second male closure element. Further, a plurality of perforations can be arranged within the first closure gap and a plurality of perforation gaps can be between adjacent perforations of the plurality of perforations wherein the plurality of perforation gaps can vary in size.

[0016] For various embodiments of the present invention, the plurality of perforations can also vary in size. Also, a first perforation of the plurality of perforations can be proximate to the first end of the first flange and a last perforation of the plurality of perforations can be proximate to the second end of the first flange. One or more central perforations of the plurality of perforations can be between the first and last perforations and a plurality of first side perforations of the plurality of perforations can be distributed between the first perforation and the one or more central perforations. In addition, the plurality of first side perforations can increase in size as the perforations move away from the first side perforation and towards the one or more central perforations.

[0017] It is contemplated that the present invention may be utilized in ways that are not fully described or set forth herein. The present invention is intended to encompass these additional uses to the extent such uses are not contradicted by the appended claims. Therefore, the present invention should be given the broadest reasonable interpretation in view of the present disclosure, the accompanying figures, and the appended claims.

Description

BRIEF DESCRIPTION OF THE RELATED DRAWINGS

[0018] A full and complete understanding of the present invention may be obtained by reference to the detailed description of the present invention and the preferred embodiments when viewed with reference to the accompanying drawings. The drawings can be briefly described as follows.

[0019] FIG. **1** provides a plan view of a fastener assembly **100** as contemplated by one embodiment of the present invention.

[0020] FIG. **2** provides a cross-section of the FIG. **1** fastener assembly taken at a center of an individual perforation of the plurality of perforations shown of FIG. **1**.

[0021] FIGS. **3**A and **3**B provide plan views of a first flange of second and third embodiments of the present invention.

[0022] FIGS. **3**C and **3**D provide plan views of a first flange of fourth and fifth embodiments of the present invention.

[0023] FIG. **4**A provides a perspective view of fastener assembly **100** applied to reclosable bag **200**.

[0024] FIG. **4**B provides a front view of the fastener assembly **100** applied to reclosable bag **200**.

[0025] FIG. **5**A provides a partial plan view of a flange showing details of a perforation of the present invention.

[0026] FIG. **5**B provides a cross-sectional view of FIG. **5**A along cutting plane E-E.
DETAILED DESCRIPTION OF THE INVENTION
[0027] The present disclosure illustrates one or more preferred embodiments of the present invention. It is not intended to provide an illustration or encompass all embodiments contemplated by the present invention. In view of the disclosure of the present invention contained herein, a person having ordinary skill in the art will recognize that innumerable modifications and insubstantial changes may be incorporated or otherwise included within the present invention without diverging from the spirit of the invention. Therefore, it is understood that the present invention is not limited to those embodiments disclosed herein. The appended claims are intended to more fully and accurately encompass the invention to the fullest extent possible, but it is fully appreciated that certain limitations on the use of particular terms is not intended to conclusively limit the scope of protection.
[0028] FIGS. **1** and **2** depict the layout and structure, respectively, of fastener assembly **100**. Fastener assembly **100**, in at least certain embodiments, can be adapted for closing and opening a reclosable bag, as shown in FIGS. **4**A and **4**B and described below.
[0029] Fastener assembly **100** includes opposing first and second flanges **102** and **104**. First flange **102** includes opposing first and second surfaces **106** and **108**. Second flange **104** includes opposing first and second surfaces **114** and **116**. As illustrated by FIG. **1**, first flange **102** further comprises opposing first and second ends **110** and **112** and second flange **104** comprises opposing first and second ends **118** and **120**. As further illustrated by FIGS. **1** and **2**, first flange **102** includes top and bottom sides **122** and **124** and second flange **104** includes top and bottom sides **126** and **128**.
[0030] FIG. **2** further illustrates upper female and lower male closure elements **123** and **127** extending distally from first flange **102**. Likewise, FIG. **2** shows upper male and lower female closure elements **125** and **129** extending distally from second flange **104**. Upper female closure element **123** interlocks with upper male closure element **125** and lower female closure element **127** interlocks with lower male closure element **129**.
[0031] As shown by FIG. **2**, upper female closure element **123** further comprises a base with upper and lower sides **132** and **140**. Upper male closure element **125** comprises a base with upper and lower sides **130** and **139**. Lower male closure element **127** comprises a base with upper and lower sides **144** and **136**. Lower female closure element **129** comprises a base with upper and lower sides **142** and **138**. Regarding the illustration of closure elements **123**, **125**, **127**, and **129**, various features of these elements are not shown by FIG. **1** for ease of illustration.
[0032] In between the lower side **140** of the base of upper female closure element **123** and upper side **144** of the base of lower male closure element **127**, a first gap A is shown by FIG. **2**. In addition, a second gap B is shown between the lower side **139** of the base of upper male closure element **125** and the upper side **142** of the base of lower female closure element **129**. In at least certain embodiments, gaps A and B can be constant for the entire length of the closure elements **123**, **125**, **127** and **129**. In at least certain embodiments, the constant distance for gaps A and B can vary depending upon the embodiment. Still, in certain embodiments, the distance can be somewhere between 0.25 inches and 0.5 inches.
[0033] Returning to FIG. **1**, a first plurality of apertures **150** is shown in first flange **102**. In at least certain embodiments, a second plurality of apertures **151** can extend through second flange **104**. FIG. **2** illustrates a first aperture **150***s* of the first plurality of apertures **150** extending through first flange **102** from first surface **106** to second surface **108**. FIG. **2** further illustrates a first aperture **151***s* of the second plurality of apertures **151** of second flange **104** extending from first surface **114** through second surface **116**.
[0034] In at least certain embodiments, each aperture of each plurality of apertures **150** and **151** can have a generally circular shape and approximately the same diameter. Additionally, in at least certain embodiments, each plurality of apertures can be equally spaced from each other as shown in FIG. **1** and arranged along a single line. In further embodiments, apertures **150** and **151** may not be

equally or regularly spaced from each other. Further, in certain embodiments, apertures **150** and **151** may not be circular and the size of the apertures may vary from one another.

[0035] The linear arrangement of apertures or perforations **150** and **151**, as shown by FIGS. **1** and **2**, can also extend continuously from the first ends **110**/**118** to second ends **112**/**120** of flanges **102**/**104**. Furthermore, because of the size and arrangement of apertures **150** and **151**, such apertures as described and shown in this disclosure are commonly referred to as perforations or micro perforations. The plurality of apertures **150** and **151** can provide a tactile guide path for closing the pair of interlocking closures with the first interlocking closure comprising upper female and male closure elements **123** and **125** and the second closure comprising lower female and male closure elements **129** and **127**. Due to the nature of the perforations, as more fully described below, the tactile guide path provided by the invention can be much more pronounced than guide paths described in the prior art.

[0036] The size of the plurality of aperture or perforations **150** and **151** can vary. For instance, in at least one embodiment, the perforations **150** and **151** can be circular and have a diameter of about 100 microns. This size prevents liquid water from passing through the perforations while being large enough to be perceived tactically while closing the fastener assembly **100**. In various other embodiments, the size of each perforation can be up to 2000 microns. In the figures, perforations **150** and **151** are not shown to scale but rather enlarged for illustration purposes.

[0037] Now examining FIGS. **3**A and **3**B, the plurality of perforations **150** can further comprise a first perforation **150***a* proximate to the first end **110** of the first flange **102** and a last perforation **150***b* proximate to the second end **112** of first flange **102**. Located centrally between first and last perforations **150***a* and **150***b*, one or more central perforations **150***c* are shown. Additionally, located between the first perforation **150***a* and the one or more central perforations **150***c*, a plurality of first side perforations **150***d* are shown by FIGS. **3**A and **3**B. Likewise, FIGS. **3**A and **3**B show a plurality of second side perforations **150***e* located between the one or more central perforations **150***c* and the last perforation **150***b*. Although only described for first flange **102**, these same features and the additional features described below can apply as well to second flange **104**.

[0038] In contrast to the FIG. **1**, wherein each perforation **150** has the same size, the size of perforations **150** of FIGS. **3**A and **3**B vary. For FIG. **3**A, first and last perforations **150***a* and **150***b* are the smallest and the one or more central perforations **150***c* are the largest. In the FIG. **3**A embodiment, the size of the apertures or perforations can increase incrementally as the perforations move away from the outwardly located first and last apertures **150***a* and **150***b* and towards the one or more central perforations **150***c*.

[0039] In contrast to FIG. **3**A, the relationship in the size and location of the perforations **150** of FIG. **3**B are reversed. The one or more central perforations **150***c* are the smallest while the opposing first and last perforations **150***a* and **150***b* are the largest. As the perforations move away from the one or more central perforations **150***c* and towards the outer first and last perforations **150***a* and **150***b*, each perforation increases in size incrementally until the outer perforations **150***a* and **150***b* are reached.

[0040] These various patterns in the size of perforations **150** can provide further tactile guidance to a person closing fastener assembly **100**. The described patterns can indicate to a user the preferred method of closing fastener assembly **100**—that is by applying pressure beginning at one end of the closure **100** (such as first flange ends **110**/**118**), moving towards the central area, and then towards the opposing end of the closure assembly **100** (such as second flange ends **112**/**120**).

[0041] Now examining FIGS. **3**C and **3**D, unlike FIGS. **3**A and **3**B, the diameters of the plurality of apertures **150** in FIGS. **3**C and **3**D remain constant; however, the distance separating adjacent apertures varies. For instance, FIGS. **3**C and **3**B illustrate a plurality of first side perforation gaps **152***a* between adjacent perforations of the plurality of the first side perforations **150***d* and a plurality of second side perforations **152***b* between adjacent perforations of the plurality of the second side perforations **150***e*.

[0042] As further illustrated by FIGS. **3**C and **3**D, the gaps between adjacent perforations **150** can vary depending upon their location along flange **102**. As shown in FIGS. **3**C, the gaps decrease as the gaps move away from the first and second ends **110** and **112** and inwards toward the center of flange **102**. In particular, the first side perforation gaps **152***a* decrease as the gaps move away from the first perforation **150***a* until the one or more central perforations **150***c* are reached. Likewise, the second side perforation gaps **152***b* decrease as the gaps move away from the last perforation **150***b* until the one or more central perforations **150***c* are reached.
[0043] The relationship between the location of gaps **152** and the size of individual gaps can also be reversed as illustrated by FIG. **3**D. The figure shows the first side perforation gaps **152***a* increasing as the gaps move away from the first perforation **150***a* until the one or more central perforations **150***c* are reached. Likewise, the figure shows the second side perforation gaps **152***b* increasing as the gaps move away from the last perforation **150***b* until the one or more central perforations **150***c* are reached. Although the size of each of the plurality of perforations **150** in FIGS. **3**C and **3**D are constant, the size of the perforations can also vary based upon their location on first flange **102**. In at certain embodiments, these above-described patterns in gap size can also be applied to second flange **104**.
[0044] The change in gap sizes as described by FIGS. **3**C and **3**D can aid in providing tactile guidance in closing fastener assembly **100**. In a similar manner as the change in perforation size disclosed by FIGS. **3**A and **3**B, the various patterns in gap size can indicate to a user the preferred method of closing the fastener assembly **100**.
[0045] FIGS. **4**A and **4**B depict a reclosable storage bag **200** utilizing reclosable fastener assembly **100**. Bag **200** is shown having a first panel **202** and a second panel **204**. The two panels **202** and **204** are joined together at a first side **206**, at a second side **208**, and at a bottom edge **210**. The first panel **202** and second panel **204** may be formed from a single piece of polymeric film which is folded to form the bottom edge **210**. Heat seals can also be used to affix panels **202** and **204** at sides **206** and **208**. The figures further show a first top edge **212** and second top edge **214** at an upper distal end of the first panel **202** and the second panel **204**, respectively. A bag opening is defined between first and second top edges **212** and **214**.
[0046] Still examining FIGS. **4**A and **4**B, a first flange **102** of closure assembly **100** is shown disposed on an interior surface of the first panel **202** and the second flange **104** of fastener assembly **100** is shown disposed on an interior surface of the second panel **204**. As shown in these figures, the two cooperating flanges **102** and **104** of closure **100** are separated from each other. Further shown by the figures, the first and second flanges **102** and **104** are shown extending from the first side **206** to the second side **208** of bag **200**. As addressed above, the first and second flanges **102** and **104** are configured to engage with each other, thereby enabling the closing and reopening of bag **200** below flanges **102** and **104**. Further detail of closure assembly **100** is not shown in FIGS. **4**A and **4**B for ease of illustration.
[0047] In certain embodiments, flanges **102** and **104** may be affixed to bag **200** by one or more seals with each seal extending in a lengthwise direction of flanges **102** and **104**. In certain embodiments, the attaching seals may be formed by heat or ultrasonically. In an alternative embodiment, flanges **102** and **104** may be affixed to bag **200** by a pressure sensitive adhesive. In further embodiments, first flange **102** can be integral with first panel **202** and second flange **102** can be integral with second panel **204**.
[0048] FIGS. **5**A and **5**B illustrate details of a certain embodiment of the plurality of perforations **105**. As shown by cross-section FIG. **5**B, second surface curves inward as the surface approaches single perforation or aperture **105***s*. FIG. **5**B further illustrates that second surface **108** of flange **102** curves upward as perforation **105***s* is approached. Additionally, FIG. **5**B shows annular-shaped aperture wall **107** extending at 90 degrees from a plane of first flange **102** with first surface **106** transitioning from horizontal to vertical as surface **106** approaches perforation **105***s*. FIG. **5**B further shows that distal end **109** of wall **107** is jagged and irregular.

[0049] Wall **107** can be formed by the formation of aperture **105***s*. Wall **107**, along with walls corresponding to each aperture of perforation of the above-described plurality of perforations **105**, can be intentionally formed during the perforation making process to increase the tactile feedback provided by the plurality of perforations **105**. At least in one embodiment, perforation **105***s* of FIG. **6** can be formed by cold dull perforating pins. The irregular features of distal end **109** of aperture wall **107** can be formed from the tearing of flange **102** when perforation **105***s* is formed.
[0050] Unlike the FIGS. **5**A and **5**B embodiment, in other embodiments, the plurality of perforations **150** may be formed by other methods, such as heated pins that do not form such a pronounced and jagged raised feature as annular wall **107**. Formation by heated pins may form an annular bulge about perforation **105***s*, but not a jagged wall as described and shown in FIG. **5**B.
[0051] As previously noted, the specific embodiments depicted herein are not intended to limit the scope of the present invention. Indeed, it is contemplated that any number of different embodiments may be utilized without diverging from the spirit of the invention. Therefore, the appended claims are intended to more fully encompass the full scope of the present invention.

Claims

1. A fastener, comprising: a first flange and an opposing second flange, first male and female closure elements extending distally from a first surface of the first flange, a first closure gap between the first male and female closure elements, second male and female closure elements extending distally from a first surface of the second flange, the first male closure element interlocking with the second female closure element, the first female closure element interlocking with the second male closure element, and a plurality of perforations arranged within the first closure gap.
2. The fastener of claim 1, further comprising: the first flange extending from a first end to a second end, the first male and female closure elements extending between the first and second ends of the first flange, and the plurality of perforations extending between the first and second ends of the first flange.
3. The fastener of claim 2, further comprising: the plurality of perforations arranged along a single line.
4. The plurality of perforations of claim 3, further comprising: a first perforation of the plurality of perforations proximate to the first end of the first flange, a last perforation of the plurality of perforations proximate to the second end of the first flange, and one or more central perforations of the plurality of perforations located between the first and last perforations.
5. The fastener of claim 4, further comprising: the one or more central perforations larger than the first and last perforations.
6. The fastener of claim 4, further comprising: the first and last perforations larger than the one or more central perforations.
7. The fastener of claim 4, further comprising: a plurality of first side perforations of the plurality of perforations distributed between the first perforation and the one or more central perforations, and a plurality of second side perforations of the plurality of perforations distributed between the last perforation and the one or more central perforations, a plurality of first side perforation gaps comprising gaps between adjacent perforations of the plurality of first side perforations, and a plurality of second side perforation gaps comprising gaps between adjacent perforations of the plurality of second side perforations.
8. The fastener of claim 7, further comprising: each of the plurality of first side perforation gaps increasing as the gaps move further away from the first side perforation and towards the one or more central perforations, and each of the plurality of second side perforation gaps increasing as the gaps move further away from the last side perforation and towards the one or more central perforations.

9. The fastener of claim 7, further comprising: each of the plurality of first side perforation gaps decreasing as the gaps move further away from the first side perforation and towards the one or more central perforations, and each of the plurality of second side perforation gaps decreasing as the gaps move further away from the last side perforation and towards the one or more central perforations.

10. The fastener of claim 7, further comprising: each of the plurality of first side perforations increasing in size as the perforations move away from the first side perforation and towards the one or more central perforations, and each of the plurality of second side perforations increasing in size as the perforations move away from the last side perforation and towards the one or more central perforations.

11. The fastener of claim 7, further comprising: each of the plurality of first side perforations decreasing in size as the perforations move further away from the first side perforation and towards the one or more central perforations, and each of the plurality of second side perforations decreasing in size as the perforations move further away from the last side perforation and towards the one or more central perforations.

12. The fastener of claim 1, further comprising: a second surface of the first flange curving inwards about an individual perforation of the plurality of perforations.

13. The fastener of claim 12, further comprising: a first surface of the first flange having an annular shaped wall surrounding an individual perforation of the plurality of perforations.

14. A fastener, comprising: a first flange and an opposing second flange, first male and female closure elements extending distally from a first surface of the first flange, a first closure gap between the first male and female closure elements, second male and female closure elements extending distally from a first surface of the second flange, the first male closure element interlocking with the second female closure element, the first female closure element interlocking with the second male closure element, a plurality of perforations arranged within the first closure gap, and the plurality of perforations varying in size.

15. The fastener of claim 14, further comprising: a plurality of perforation gaps comprising gaps between adjacent perforations of the plurality of perforations, and the plurality of perforation gaps varying in size.

16. The fastener of claim 14, further comprising: one or more central perforations of the plurality of perforations located between opposing first and second ends of the first flange, a plurality of first side perforations of the plurality of perforations distributed between the first end of the first flange and the one or more central perforations, and the plurality of first side perforations varying in size from the one or more central perforations.

17. The fastener of claim 14, further comprising: a first perforation of the plurality of first side perforations proximate to a first end of the first flange, a last perforation of the plurality of perforations proximate to a second end of the first flange, one or more central perforations of the plurality of perforations located between the first and last perforations, a plurality of first side perforations of the plurality of perforations distributed between the first perforation and the one or more central perforations, and the plurality of first side perforations increasing in size as the perforations move further away from the first side perforation and towards the one or more central perforations.

18. A fastener, comprising: a first flange and an opposing second flange, first male and female closure elements extending distally from a first surface of the first flange, a first closure gap between the first male and female closure elements, second male and female closure elements extending distally from a first surface of the second flange, the first male closure element interlocking with the second female closure element, the first female closure element interlocking with the second male closure element, a plurality of perforations arranged within the first closure gap, a plurality of perforation gaps comprising gaps between adjacent perforations of the plurality of perforations, and the plurality of perforation gaps varying in size.

19. The fastener of claim 18, further comprising: the plurality of perforations varying in size.
20. The fastener of claim 18, further comprising: a first perforation of the plurality of perforations proximate to the first end of the first flange, a last perforation of the plurality of perforations proximate to the second end of the first flange, one or more central perforations of the plurality of perforations between the first and last perforations, a plurality of first side perforations of the plurality of perforations distributed between the first perforation and the one or more central perforations, and the plurality of first side perforations increasing in size as the perforations move away from the first side perforation.